

Trees

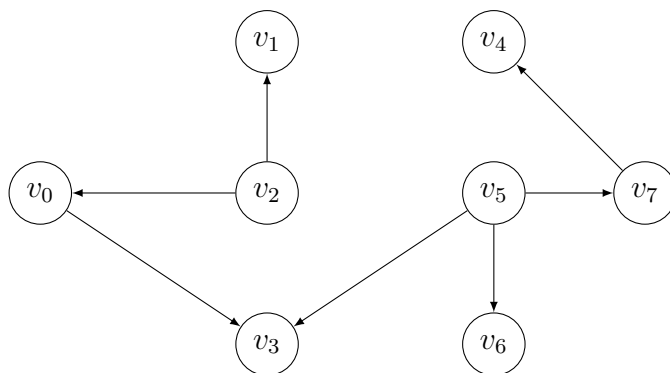
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Solution of the practice quiz

First, recall that given a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \Phi)$, a subgraph $(\mathcal{V}, \mathcal{E}', \Phi')$ is a spanning tree of $\mathcal{G}$ if:

- $\mathcal{E}' \subseteq \mathcal{E}$,
- $\Phi'(e) = \Phi(e)$ for each $e \in \mathcal{E}'$,
- $(\mathcal{V}, \mathcal{E}', \Phi')$ is a tree.

An example of spanning tree is the following:



It is easily verified that it is a subgraph involving all the nodes. Furthermore, it is a connected graph without cycles and thus a tree.